<Online Reviews trustworthiness score > - <Adel Dghim>

Project Context

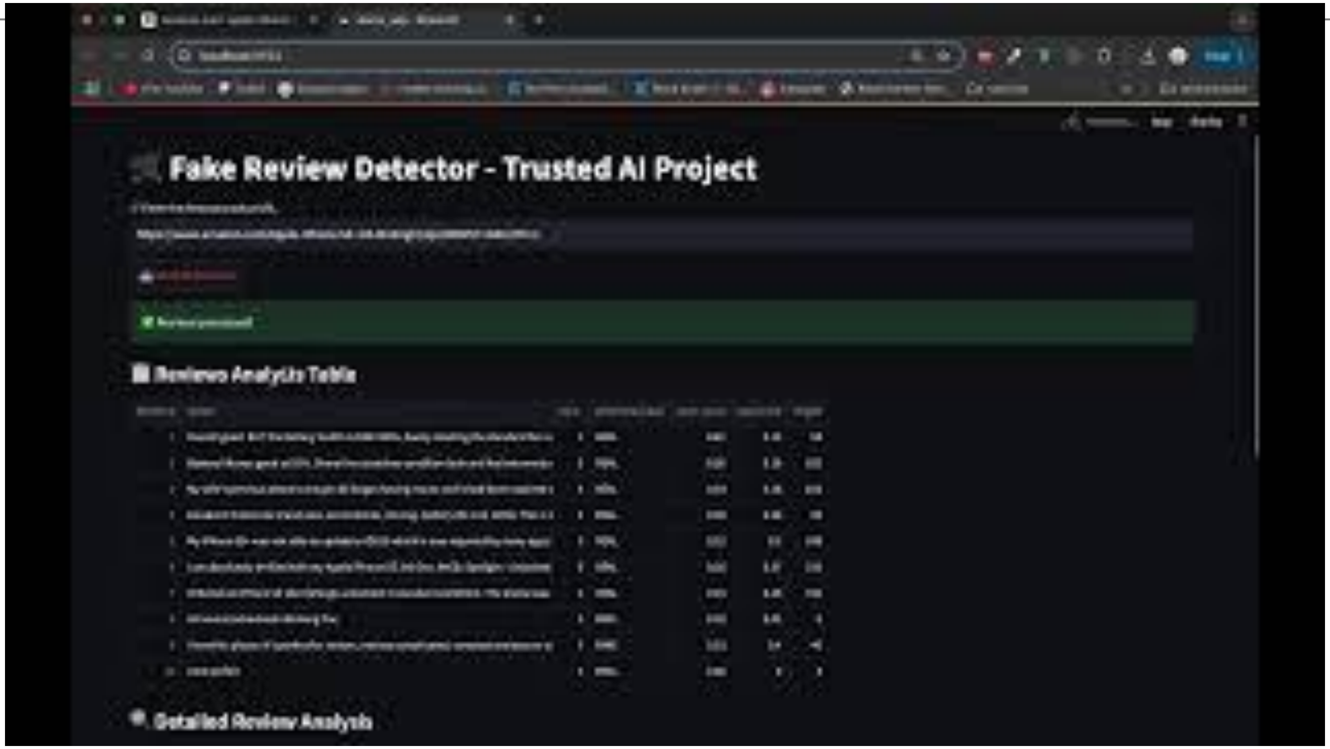
1. Title:Online Reviews trustworthiness score
2. Key idea: (2-3 lines): assigning a **trust score** (from 0 to 1) to online reviews using semantic understanding (BERT), metadata (stars, length, sentiment), and explainability with SHAP.
3. Who will care when done: Consumers, product managers, platform moderators…
4. Data need:d iverse and realistic dataset of real and fake reviews from multiple sources (Amazon, Yelp, LLMs, Ott dataset…)
5. Methods: Sentence-BERT (MiniLM) + Random Forest, Trust Score (0 to 1), Metadata, SHAP
6. Evaluation:

	precision	recall	f1-score	support
real	0.87	0.99	0.92	3301
fake	0.85	0.26	0.40	690
accuracy			0.86	3991
macro avg	0.86	0.63	0.66	3991
weighted avg	0.86	0.86	0.83	3991

AUC ROC: 0.838

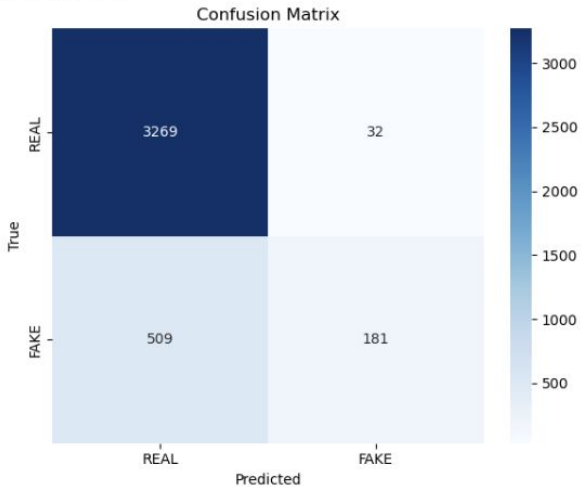
7. Users: End-users reading reviews / Platforms filtering content
8. Trust issue: realistic-looking reviews can be fake. LLMs make deception easier. But fake-looking, over-generic or very short reviews can also be real.

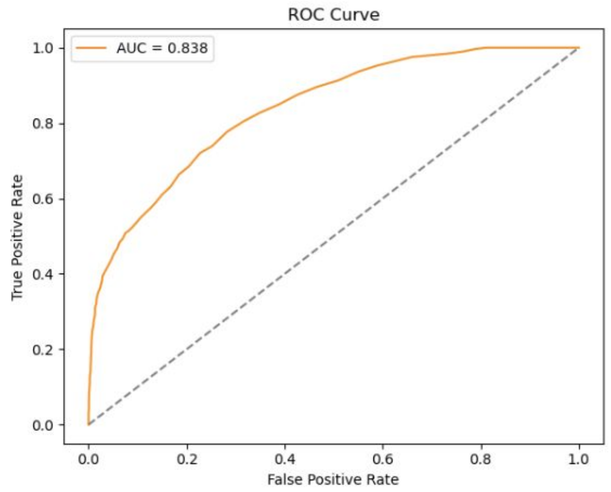
DEMO


Fake Review Detector - Trusted AI Project
Reviews Analytics Table
Detailed Review Analysis

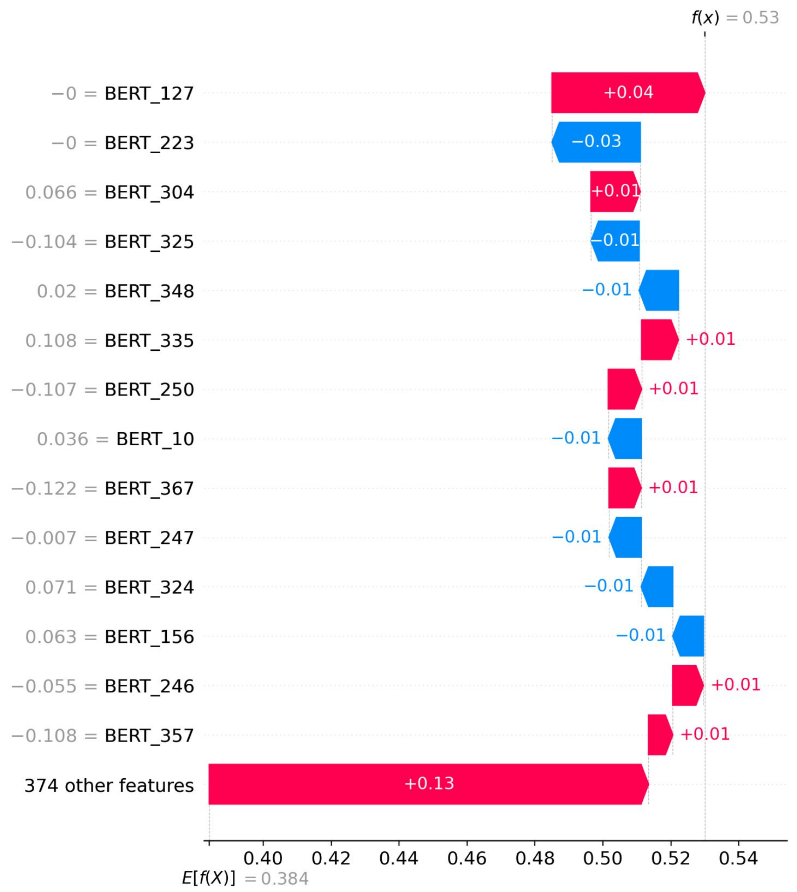
Metrics:

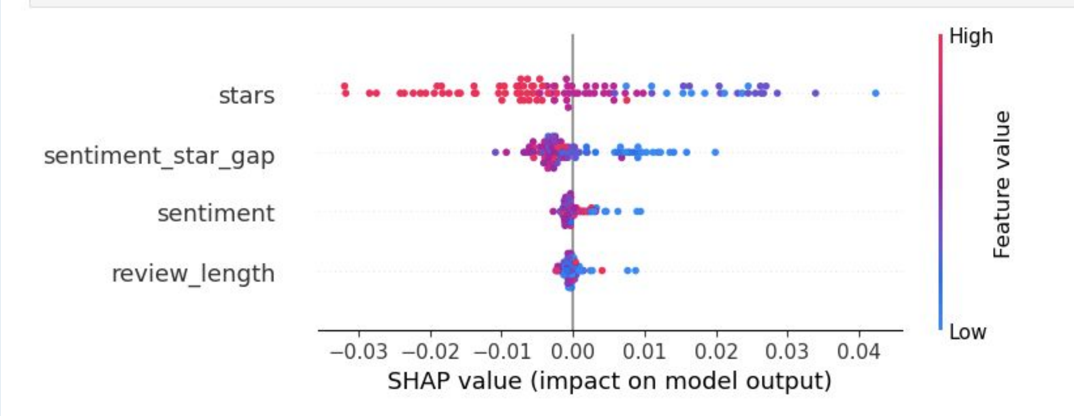
	precision	recall	f1-score	support
real	0.87	0.99	0.92	3301
fake	0.85	0.26	0.40	690
accuracy			0.86	3991
macro avg	0.86	0.63	0.66	3991
weighted avg	0.86	0.86	0.83	3991





Explainability:





Comparison with Grok 3

Prompt: You are a fake review detection assistant trained to assess the trustworthiness of customer reviews.

Your task:

1. Analyze the following customer review.

2. Give a **confidence score between 0 and 1** representing how likely it is that the review is **fake**. - 0 = certainly real - 1 = certainly fake

3. Justify your reasoning based on **writing style, sentiment-to-stars mismatch, length, specificity**, or **typical patterns of deceptive content**.

Review metadata: - Category: Electronics - Stars: 5/5 Customer Review: "I love this phone. It's perfect for seniors, not too complicated, compact and easy to use. It's a fairly recent model (still some warranty left), battery is 100% and the price was great. I'm very happy I found such a great refurbished phone on Amazon."

Your response format should be: Prediction: [FAKE or REAL] Trust Score (Fake): [e.g. 0.72] Reasoning

Output:
Prediction: REAL
Trust Score (Fake): 0.25
Reasoning: The review exhibits characteristics of a genuine review. The writing style is conversational and natural, avoiding overly polished or promotional language typical of fake reviews. The specificity is moderate, mentioning details like "perfect for seniors," "compact," "recent model," "warranty left," and "battery is 100%," which suggest actual user experience rather than vague superlatives. The sentiment aligns well with the 5/5 stars, expressing consistent positivity without exaggerated claims. The length is concise yet informative, typical of authentic reviews rather than overly brief or excessively detailed fake ones. However, the mention of "refurbished phone on Amazon" and the price being "great" slightly raises suspicion, as fake reviews sometimes emphasize deals to promote sales. This minor red flag prevents a score of 0, but overall, the review lacks strong deceptive patterns.

Conclusion

Challenges:

- **Building a realistic and trustworthy dataset**

- **Ambiguity in Ground Truth :** Defining what constitutes a *fake* review is complex

  YelpZIP and Amazon contains labels based on filtering that might not be 100% accurate.
  LLM-generated reviews might not be detectable as fake.
  Fake reviews dataset with Computer generated reviews has low realism.

- **Bias Toward Linguistic Simplicity**

  The model may inadvertently penalize:

  overly enthusiastic or emotionally charged reviews
  short or generic phrasing
  reviews with mismatched sentiment and rating

  These behaviors may be present in legitimate human reviews

- **Explainability**

  semantic embedding dimensions (BERT_123, etc.) are not human interpretable, making it difficult to explain *why* certain textual aspects contribute to a "fake" label.